

CHARTS & GRAPHS



In this section we'll cover visualizing data with **charts & graphs**, including how to identify the right chart type, how to format it properly, and how to use it to tell a story

TOPICS WE'LL COVER:

Data Viz 101

Creating Charts

Essential Visuals

GOALS FOR THIS SECTION:

- Understand the importance of visualizing data to communicate numbers effectively
- Learn to choose the right chart type
- Create and customize basic charts in Excel, including line, bar, pie charts, and more!

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WHY VISUALIZE DATA?

Data visualization allows you to bring your data to life

- The human brain is built to interpret raw data as meaningless numbers and noise
- We need **clear patterns** and **visual cues** to help us quickly make sense of complex information

In 10 seconds, what can you learn from the data below?

Product Sales by Month (in millions)

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Product A	4.8	5.78	6.24	6.34	6.95	3.02	8.45	8.79	10.3	9.93	11.4	11.56
Product B	0.67	1.05	1.62	2.67	3.91	5.49	8.36	10.99	13.58	14.81	15.13	15.26
Product C	4.53	4.61	4.74	5.1	5.32	5.7	5.77	6.32	6.56	6.64	18.5	19.8
Product D	8.35	7.72	12.05	7.7	7.05	11.05	6.95	6.39	9.5	4.83	4.03	8.03

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WHY VISUALIZE DATA?

Data Viz 101

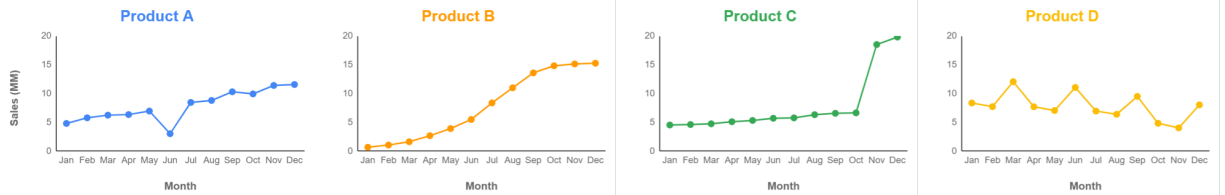
Creating Charts

Essential Visuals

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*What if you **visualize** it?*

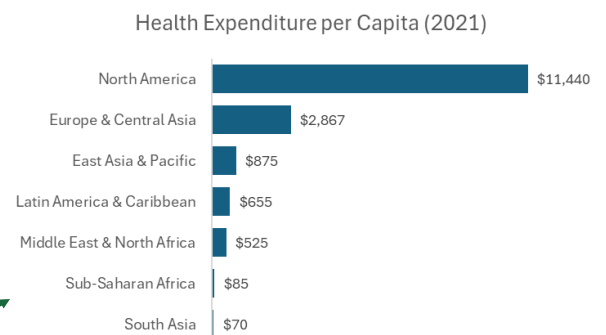
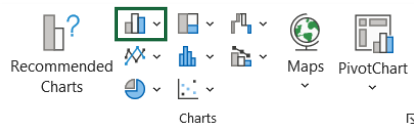


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CREATING CHARTS

You can **create a chart** in Excel by selecting a chart type from the Insert tab:

Region	Health Expenditure per Capita
North America	\$11,440
Europe & Central Asia	\$2,867
East Asia & Pacific	\$875
Latin America & Caribbean	\$655
Middle East & North Africa	\$525
Sub-Saharan Africa	\$85
South Asia	\$70



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CHOOSING THE RIGHT CHART TYPE

When **choosing the right chart type**, there are two main things to focus on:

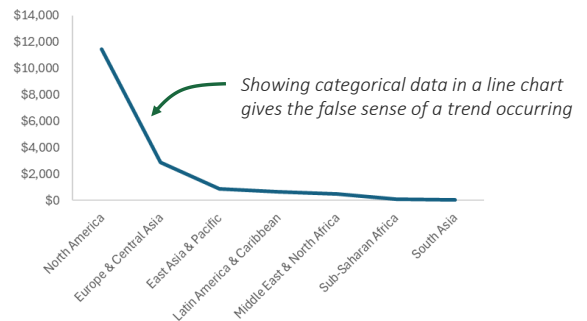
- The **type of data** you're working with (*categorical, time series, or numerical*)
- What you want to **communicate** (*comparison, composition, relationship, or distribution*)

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North America	\$11,440
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We want to **compare** health expenditure across these regions, or **categories**



Health Expenditure per Capita (2021)



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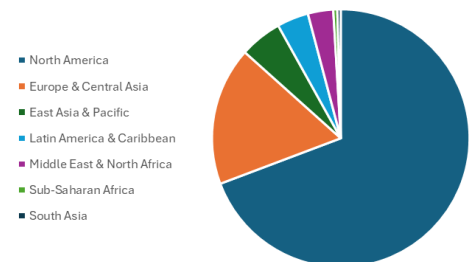
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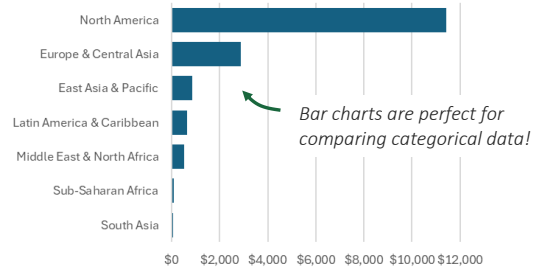
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Health Expenditure per Capita (2021)



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LINE CHARTS

Line charts are commonly used for visualizing trends over time

- They use **time series** data and communicate **comparison**

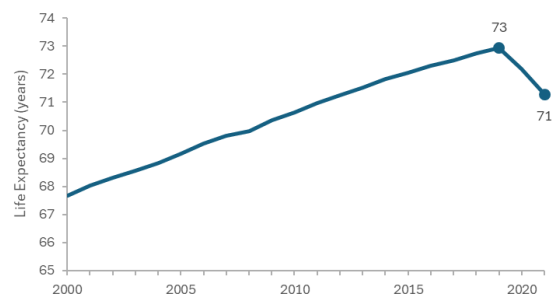
Column for each line

Year	Life Expectancy
2000	68
2001	68
2002	68
2003	69
2004	69
2005	69
2006	70
2007	70
2008	70
2009	70

Continuous dates in the rows



Global life expectancy dropped during COVID



PRO TIP: If you have multiple series, try to highlight one or two lines using color and thickness and mute out the rest to focus on a clear story

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STACKED AREA CHARTS

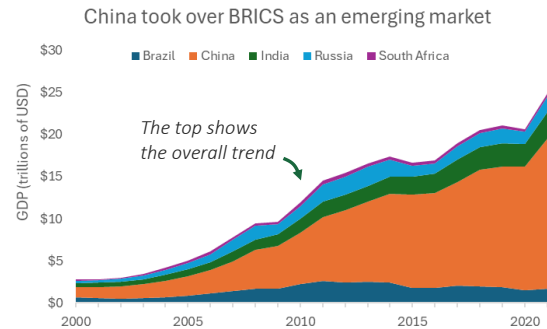
Stacked area charts are used for visualizing changes in composition over time

- They use **time series** data and communicate both **comparison** and **composition**

Continuous dates in the rows

Column for each area

Year	Brazil	China	India	Russia	South Africa
2000	\$0.7	\$1.2	\$0.5	\$0.3	\$0.2
2001	\$0.6	\$1.3	\$0.5	\$0.3	\$0.1
2002	\$0.5	\$1.5	\$0.5	\$0.3	\$0.1
2003	\$0.6	\$1.7	\$0.6	\$0.4	\$0.2
2004	\$0.7	\$2.0	\$0.7	\$0.6	\$0.3
2005	\$0.9	\$2.3	\$0.8	\$0.8	\$0.3
2006	\$1.1	\$2.8	\$0.9	\$1.0	\$0.3
2007	\$1.4	\$3.6	\$1.2	\$1.3	\$0.3
2008	\$1.7	\$4.6	\$1.2	\$1.7	\$0.3
2009	\$1.7	\$5.1	\$1.3	\$1.2	\$0.3



PRO TIP: Use a **100% stacked area chart** to focus on the changes in composition (by losing the comparison)

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STACKED AREA CHARTS

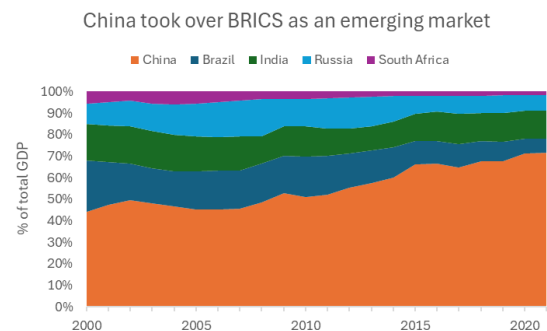
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2003	\$0.6	\$1.7	\$0.6	\$0.4	\$0.2
2004	\$0.7	\$2.0	\$0.7	\$0.6	\$0.3
2005	\$0.9	\$2.3	\$0.8	\$0.8	\$0.3
2006	\$1.1	\$2.8	\$0.9	\$1.0	\$0.3
2007	\$1.4	\$3.6	\$1.2	\$1.3	\$0.3
2008	\$1.7	\$4.6	\$1.2	\$1.7	\$0.3
2009	\$1.7	\$5.1	\$1.3	\$1.2	\$0.3



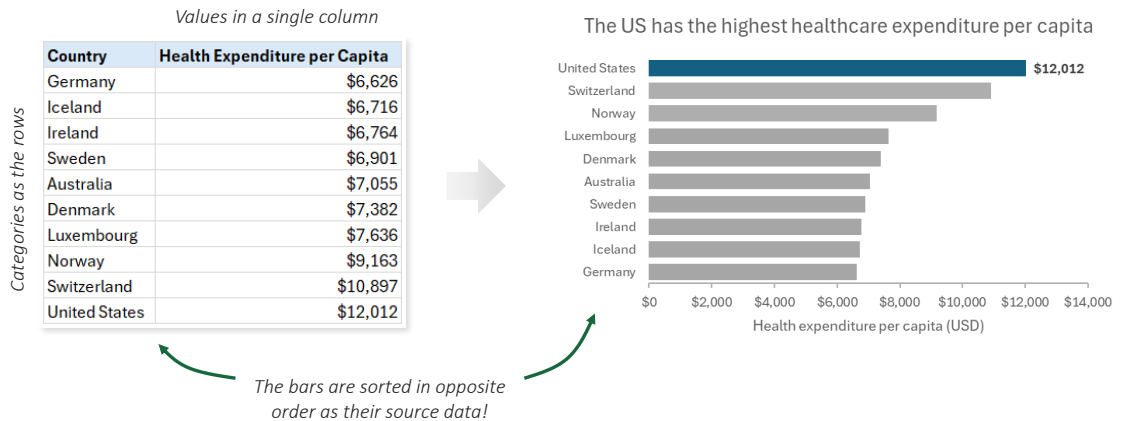
PRO TIP: Use a **100% stacked area chart** to focus on the changes in composition (by losing the comparison)

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BAR CHARTS

Bar charts are used for comparing values across different categories

- They use **categorical** data and communicate **comparison**

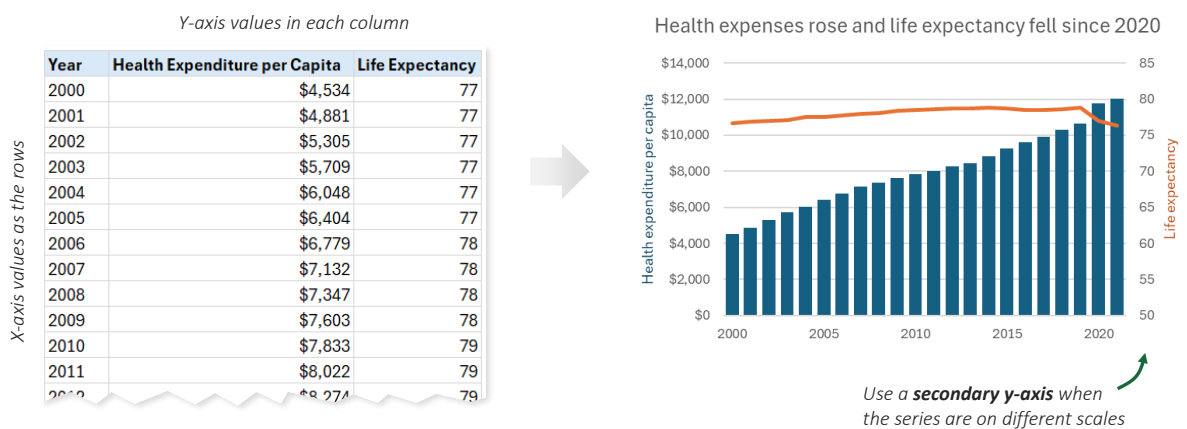


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COMBO CHARTS

Combo charts are used for visualizing two or more series of data using multiple chart types within a single visual

- They typically use a combination of a line and column chart
- They work best with **time series** data to communicate **comparison** and **relationships**

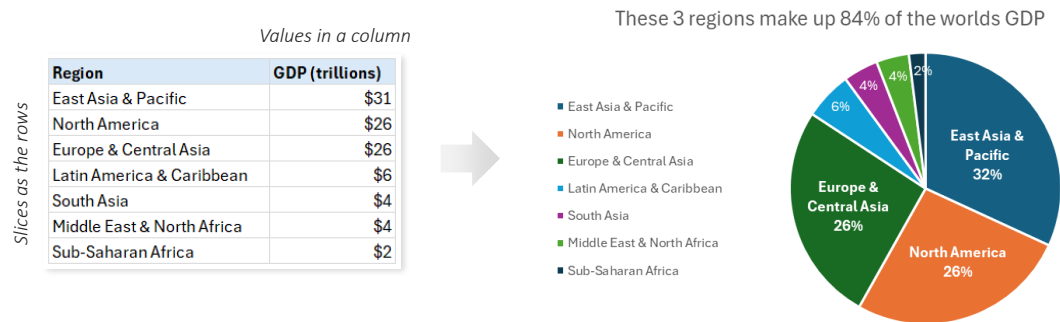


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PIE & DONUT CHARTS

Pie & donut charts are used for visualizing part-to-whole relationships

- They use **categorical** data to communicate **composition**



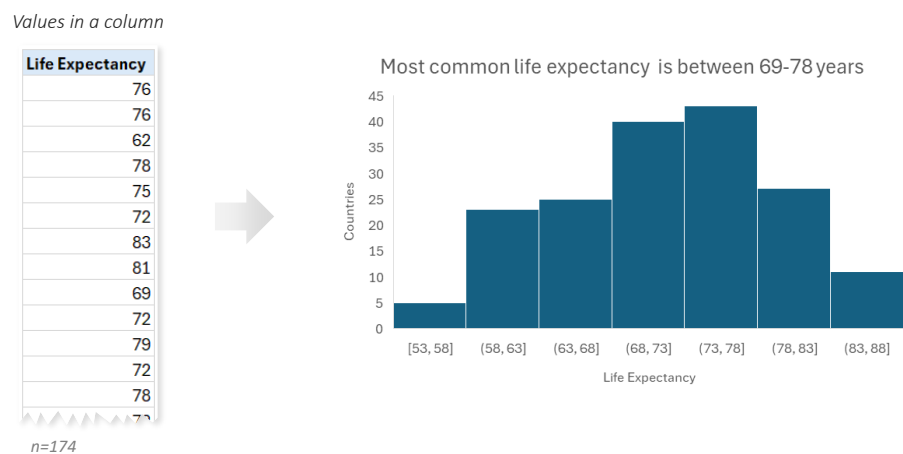
PRO TIP: Keep the number of slices low (<6) to maximize readability; you can group “others” into a single slice

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HISTOGRAMS

Histograms are used for visualizing the distribution of a numerical variable

- They use **numerical** data to communicate **distribution**



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SCATTERPLOTS

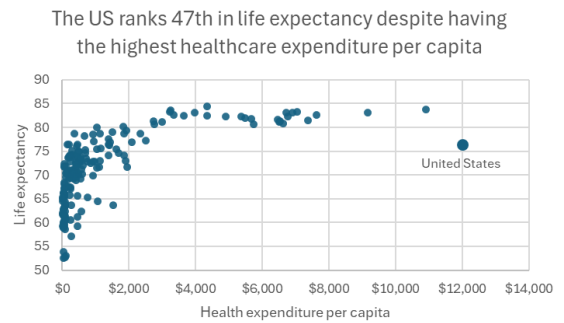
Scatterplots are used for visualizing the relationship between numerical variables

- They use **numerical** data to communicate **relationships**

One row per point

Health Expenditure per Capita	Life Expectancy
\$465	76
\$205	76
\$64	62
\$923	78
\$1,045	75
\$613	72
\$7,055	83
\$6,505	81
\$249	69
\$1,962	72
\$1,146	79
\$58	72
\$1,400	78
\$1,400	78

n=174



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MAPS

Maps are used to visualize data for geographic locations (*countries, cities, states, etc.*)

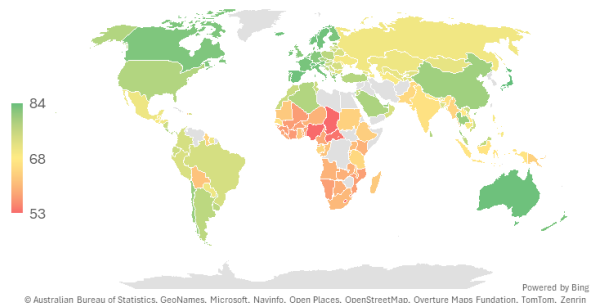
- They use **geographical** data to communicate **comparisons**

Values in a column

Country	Life Expectancy
Albania	76
Algeria	76
Angola	62
Antigua and Barbuda	78
Argentina	75
Armenia	72
Australia	83
Austria	81
Azerbaijan	69
Bahamas, The	72
Bahrain	79
Bangladesh	72
Barbados	78
Belarus	72
Belgium	82

One row per location

Life expectancy is highest in Europe and lowest in Africa



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